Neven Caplar

LINCC Frameworks @ University of Washington
Seattle, WA, USA
ncaplar@uw.edu

Web: ncaplar.com
GitHub: nevencaplar
Scholar: Google Scholar
ORCID: 0000-0003-3287-5250

Professional Summary

Scientific data systems specialist with 8+ years of experience in large-scale astronomical surveys and data infrastructure. Background includes distributed data analysis, scientific pipeline development, and technical leadership on cross-institutional software efforts. Currently serving as scientific lead for scalable catalog analysis tools, coordinating engineering and science teams to deliver reliable, production-ready systems

Experience

University of Washington / LINCC Frameworks *2022 – Present*
Research scientist/engineer

- Scientific lead and project owner of LSDB software, enabling scalable analysis and cross-matching of billion-object astronomical catalogs.
- Own technical roadmap and guide architectural decisions for large-scale distributed data analysis
- Coordinate cross-institutional development with NASA centers (Space Telescope, IPAC), international data centers, and survey teams.
- Manage a team of software engineers and scientists; lead planning and delivery.

Vera C. Rubin Observatory (LSST) *2022 – Present*
Project Scientist

- Contribute to commissioning and validation of LSST time-domain pipelines, including difference imaging and large-scale lightcurve construction.
- Define scientific requirements and validation strategies for pipelines processing nightly multi-terabyte data volumes.

Princeton University *2017 – 2022*
Associate Professional Specialist

- Developed calibration algorithms for the Prime Focus Spectrograph (Subaru Telescope).
- Applied physical modeling and machine-learning methods to improve calibration fidelity.

Education

Ph.D., Science, Swiss Federal Institute of Technogloy, Zürich 2017
M.Sc., Physics, University of Zagreb 2010

Research

- 20 peer-reviewed publications (7 first-author); ∼1,640 total citations
- Ph.D. thesis: Evolution of the AGN Population in the Universe
- Research areas: large-scale data analysis, time-domain astronomy, survey pipelines, scientific software infrastructure

Skills

Leadership: technical project leadership, roadmap ownership, cross-institutional coordination
Data & Software: Python, distributed analytics (Dask), JAX, scientific pipelines, large catalogs
Domains: time-domain surveys, data validation, scalable scientific infrastructure